Anton Perch

Senior Software Engineer | Frontend & Full-Stack

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SUMMARY

Senior Software Engineer with 9 years shipping production web, desktop and real-time applications in TypeScript, React, Next.js and Node.js/NestJS. Strong in frontend architecture, accessibility (WCAG 2.1 AA, RGAA), performance and testing, with hands-on full-stack and applied AI experience. Works in Agile product teams, taking features from idea to release and contributing to product growth. Former tech lead and mentor; solo-builds and ships commercial Steam games.

SKILLS

Core stack: TypeScript, React, Next.js, Node.js, NestJS, GraphQL

Frontend: JavaScript, Redux, Zustand, XState, HTML/CSS (SCSS), Tailwind CSS, SSR, Web Performance, Accessibility (WCAG 2.1 AA, RGAA), Storybook, Design Systems, Atomic Design, Responsive Design

Backend & APIs: REST, OpenAPI, WebSockets, PostgreSQL, MongoDB, Supabase, Firestore

Testing & DevOps: Jest, Vitest, Cypress, Testing Library, Docker, CI/CD (GitHub Actions, Jenkins), Vite, Webpack, Vercel

Security: OAuth 2.0, OpenID Connect, JWT, Session Auth, RBAC, OWASP Top 10 (XSS, CSRF, CSP)

AI & Agentic: OpenAI and Anthropic APIs, Structured Outputs, Streaming, Prompt Versioning, Braintrust, MCP Servers, Claude Code, Codex

Also worked with: Electron, Three.js (WebGL), WebCodecs, Steamworks SDK, Stripe, Symfony Twig templates

PROFESSIONAL EXPERIENCE

Senior Frontend Developer *(Aug 2024 - Present)* **ITSharkz**

Frontend for two products in Agile teams: Swizy (B2B events SaaS) and the Qobuz.com redesign (hi-res music streaming)

- Build product features for **Swizy, a B2B events and discounts SaaS**, in Next.js, React, TypeScript and GraphQL within a large monorepo of interdependent modules, including a Google Maps event explorer built from scratch with marker clustering.
- Built the component library for the **Qobuz.com redesign from scratch** following Atomic Design: SSR components on Symfony Twig templates with TypeScript Stimulus controllers, each covered by Storybook stories, integration and accessibility tests.
- Integrate the new design across all Qobuz regional domains: layouts for new pages, responsive behavior and RGAA / WCAG 2.1 AA compliance, while fixing long-standing critical issues in legacy code.
- Built a **custom MCP server automating the team's delivery pipeline** from Jira ticket to Jenkins test run (Git, Jira, Slack, Confluence), later **evolved into an organization plugin on the company's internal marketplace**; set up Claude Code and Codex for daily development with MCP servers (GitHub, Jira, Figma, Storybook), custom skills and CLAUDE.md/AGENTS.md conventions.
- Work in a fully Agile product team: take features from refinement and Figma designs to release, clarify requirements with product and design, contribute to product growth, shape API contracts with backend teams and review PRs.

Senior Software Engineer *(Sep 2021 - Aug 2024)* **Dashbouquet Development**

Full-stack on 7 client products: AI assistants, interactive adtech, Web3 and telemedicine

- **Wing AI** (AI dating assistant, iOS/Android): led the migration from serverless Firebase to NestJS, **cutting response times from 10-15s to ~3s**; architected the AI pipeline (Claude Haiku for image understanding, structured JSON handoff to GPT) with streaming, prompt versioning and Braintrust observability.
- Promoted to **Tech Lead** after 3 months: led 2 mobile developers, planned and reviewed their work, and drove roadmap decisions with the client and an external designer.
- **Outernets AI** (adtech startup, interactive digital signage), lead developer: an Electron app on Apple Silicon Macs analyzes a live camera feed (people detection, demographics, engagement) and streams it over WebSocket to remotely deployed React ad apps, 3-4 new ones a month.
- Built its client dashboard (app deployment, online status, logging, funnel and audience analytics) and rebuilt the

team after turnover, hiring a backend and 2 frontend developers and mentoring them through ramp-up.

- **Ithaca Protocol** (crypto airdrop platform): built the social integrations module linking X, Discord, Telegram and Instagram accounts via each platform's OAuth/API flows, with a quest system for airdrop bonuses; React UI and Node.js backend.
- **Telemedicine:** React, TypeScript and GraphQL UI for Health Espresso's provider communication platform; later BetterYou's investor-demo MVP with voice navigation (speech-to-text fed to GPT, flow in XState), including its architecture and tooling.
- **AI tools and R&D:** an OpenAI + HubSpot email assistant for sales teams with Braintrust prompt management (Archive); image-to-PPTX feasibility research with Tesseract OCR and vision models (ProSights).

Software Engineer *(Jul 2019 - Aug 2021)* **XB Software**

DHTMLX Suite team: a commercial, framework-agnostic JavaScript UI component library

- Built UI widgets compatible with React, Angular, Vue, Svelte and legacy environments, integrating them into client apps with demo projects and usage docs.
- Co-developed the library's **reactive virtual DOM in vanilla JavaScript**: class-based components with lifecycle hooks and inheritance for fine-grained control over component mutations, with render scheduling on the event loop and microtask queue.
- Optimized rendering performance and cross-browser support down to IE8 (Babel, polyfills), covered components with unit and e2e tests (Jest, Mocha/Chai, Cypress) and mentored 2 junior developers.

Junior Full-Stack Engineer *(Aug 2017 - Jul 2019)* **Travel Agency**

In-house full-stack development of internal business tools for a travel agency

- Built a CRM from scratch for booking management and sales automation (JavaScript, PHP, WebSockets) with role-based access and real-time activity tracking, gathering requirements directly with the business owner.

INDEPENDENT PRODUCTS

Last Ones Down | Electron, React, Three.js | In development, Steam page live

Online social-survival game for up to 16 players

- Solo-designed a **serverless, host-authoritative architecture** over Steam Lobbies and P2P with zero server cost: a pure deterministic engine (reducers, seeded PRNG, clocks and entropy banned by custom ESLint rules) lets a match continue on another machine when the host leaves.
- Built push-to-talk voice chat from scratch, as the Steam bindings expose no voice API: Opus encoding via WebCodecs, a separate voice lane in the P2P packet format and host relay.
- TypeScript monorepo with a local quality gate (Vitest unit tests, headless full-match smoke tests, visual regression over WebGL/DOM captures), developed through an **agentic Claude Code pipeline** with custom skills, an orchestrator and a blind visual-review sub-agent.

Relax Scramble | Electron, React 19, Redux Toolkit, Vite | Released on Steam

Word puzzle game with online co-op

- **Shipped and run solo end-to-end**: Steam cloud saves, achievements and rich presence, 8 languages, Jest and Testing Library coverage, and four post-launch updates driven by player feedback and sales data.
- Implemented real-time 2-player online co-op directly on the Steamworks P2P layer (session handling, reliable delivery), with crash-safe, idempotent reward grants.

Stash Loot Stack | Next.js, Supabase, Stripe, Vercel | Live

Collectibles portfolio tracker SaaS

- Built and launched solo: scheduled price sync (CSFloat API, TCG feeds) into value, P/L and ROI dashboards, Steam inventory import, a Stripe paid tier with email price alerts (Resend), Sentry and PostHog monitoring.

EDUCATION

Master of Engineering in IT & Ergonomics

Belarusian State University of Informatics and Radioelectronics (2014 - 2020)

LANGUAGES

Russian (native), English (C1), Polish (B1)